Statistical computation and analysis 2026-2
Course number: 367-1-4361

Lecturer: Opher Donchin
Tutorials: Noam Gonen

Communication with the instructors: by e-mail
Office hours: by appointment

Syllabus

An introduction to statistics for students with a strong engineering background. The course is based on a Bayesian statistical approach with a heavy emphasis on programming and algorithms, although it also covers frequentist methods in order to give students the ability to understand and apply them. It introduces the concepts of data, models, and inference. It discusses how data can be used to update models, emphasizing a modern Bayesian workflow, and how model comparison can be used to make inference. In covering Bayesian modeling, we cover algorithms including MCMC, HMC, and NUTS in some depth. In addition to basic Bayesian modeling, we show how the same methods can be used to construct more complex models including hierarchical models, regression models, ANOVA and GLMs. The course relies on Python and the PyMC package for sampling from Bayesian models and associated support packages including ArViz, Preliz, Bambi and PyTensor. Frequentist methods include maximum likelihood estimation, confidence intervals and hypothesis testing: t tests, F tests for anova and regression, chi squared tests, and the significance of the correlation coefficient.

Additional materials:

1) The course follows chapter 1-6 and 10 of the book: Martin Osvaldo A, *Bayesian Analysis with Python*. Packt Publishing. 2024. ISBN 978-1-80512-716-1

2) Lecture slides and recordings of previous lectures are available on the moodle site.

Course requirements and grading

The course grade is based on four components:

Component	Contribution to course grade
Review activity	5% + up to 2.8% extra credit
Exercises	15%
Midterm quiz	10% magen - can only improve the grade
Final exam	70%

Students must pass the final exam in order to pass the course. A full description of the grading policy is available in a separate file.

Schedule

Lecture #	Date	Lecture	Book chapters	Date	Tutorial	Comments
				17.3	Tutorial 0 - Python	
1	18.3	Data and probability	Ch 1, 1-27	24.3	Tutorial 1 - Data simulation	
2	25.3	Models and Bayesian updating	Ch 1, 28-44		*Passover*	
Passover						
		Passover	Ch 2, 48-63	14.4	Tutorial 2	**HW 1 handed out**
3	15.4	PyMC and sampling	Ch 2, 64-87	21.4	*Memorial day*	
	22.4	*Independence day*		28.4	*No tutorial - Semester 1 exam week*	
	29.4	*No lecture - Semester 1 exam week*	Ch 2, 64-87	5.5	Tutorial 3 – [HW1 WooClap]	**HW 1 due**
4	6.5	MCMC	Ch 10, 308-324, 327-338	12.5	Mid Term	
5	13.5	Hierarchical Models	Ch 4, 111-125, 133-136	19.5	Tutorial 4	

Lecture #	Date	Lecture	Book chapters	Date	Tutorial	Comments
6	20.5	Null hypothesis testing and power	Ch 3, 91-102 Ch 4, 136-145	26.5	*Id el Adha* Tutorial 5	**HW2 handed out**
7	27.5	Normal models	Ch 5, 147-171	2.6	Tutorial 6	
			Ch 6, 186-198			
8	3.6	Linear Models	Ch 6, 200-211	9.6	Tutorial 7	**HW2 due -** 13.06.26, **23:59**
9	10.6	Interactions, ANOVA, multi-variate ANOVA		16.6	Tutorial 8 [HW 2 WooClap]	**HW3 handed out**
10	17.6	Model comparison		23.6	Tutorial 9	
11	24.6	Repeated measures and mixed models, correlations		30.6	Tutorial 10 [HW 3 WooClap]	**HW 3 due**
12	01.7	Questionable research practices				